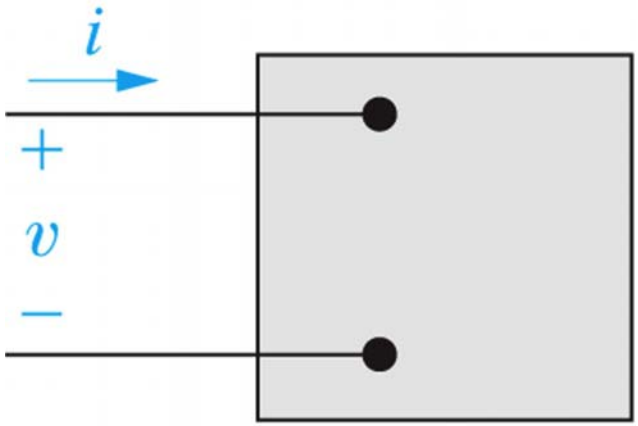
Sinusoidal Steady-State Power Calculations

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Structure

- Instantaneous Power
- Average and Reactive Power
- The rms Value and Power Calculations
- Complex Power
- Power Calculations
- Maximum Power Transfer

Instantaneous Power



$$v = V_m \cos(\omega t + \theta_v)$$
$$i = I_m \cos(\omega t + \theta_i)$$

It is convenient to use a zero time corresponding to the instant the current is passing through a positive maximum.

$$v = V_m \cos(\omega t + \theta_v - \theta_i)$$
$$i = I_m \cos \omega t$$

$$p = vi$$

$$p = V_m I_m \cos(\omega t + \theta_v - \theta_i) \cos \omega t$$

$$\cos\alpha \cos\beta = \frac{1}{2}\cos(\alpha - \beta) + \frac{1}{2}\cos(\alpha + \beta)$$

$$\alpha = \omega t + \theta_v - \theta_i$$
$$\beta = \omega t$$

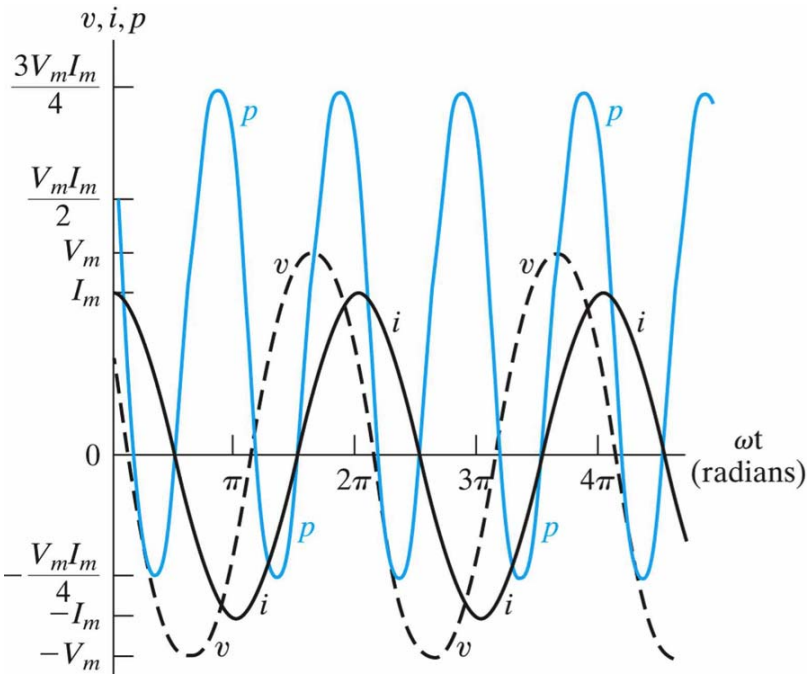
$$p = \frac{V_m I_m}{2}\cos(\theta_v - \theta_i) + \frac{V_m I_m}{2}\cos(2\omega t + \theta_v - \theta_i)$$

$$\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$$

$$p = \frac{V_m I_m}{2}\cos(\theta_v - \theta_i) + \frac{V_m I_m}{2}\cos(\theta_v - \theta_i)\cos 2\omega t - \frac{V_m I_m}{2}\sin(\theta_v - \theta_i)\sin 2\omega t$$

$\theta_v = 60°$
$\theta_i = 0°$

The frequency of the instantaneous power is twice the frequency of the voltage or current. Therefore, *the instantaneous power goes through two complete cycles for every cycle of either the voltage or the current*.



The instantaneous power may be negative for a portion of each cycle, even if the network between the terminals is passive. In a completely passive network, *negative power implies that energy stored in the inductors or capacitors is now being extracted*.

Average and Reactive Power

Eq. 10.8 can be rewritten as:

$$p = P + P\cos 2\omega t - Q\sin 2\omega t$$

If we denote

$$P = \frac{V_m I_m}{2}\cos(\theta_v - \theta_i)$$ Average (real) power

$$Q = \frac{V_m I_m}{2}\sin(\theta_v - \theta_i)$$ Reactive power

The integral of both $\cos 2\omega t$ and $\sin 2\omega t$ over one period is zero.

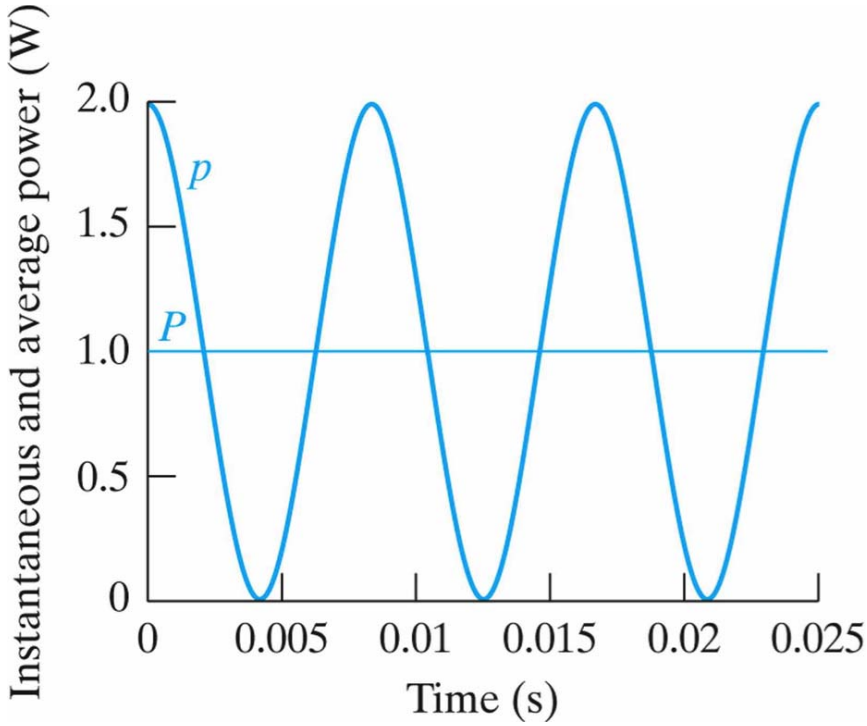
$$P = \frac{1}{T}\int_{t_0}^{t_0+T} p\,dt$$ nT

Power for purely Resistive Circuits

$$\theta_v = \theta_i \Rightarrow Q = 0$$

$$p = P + P\cos 2\omega t$$

Instantaneous real power



The instantaneous real power can never be negative, which is also shown in the figure. In other words, *power cannot be extracted from a purely resistive network. Rather, all the electric energy is dissipated in the form of thermal energy*.

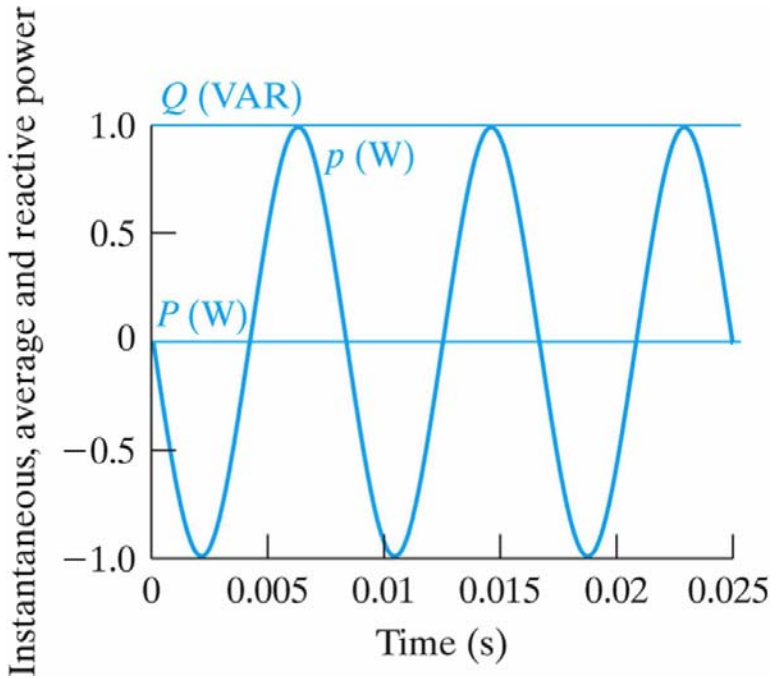
Power for Purely Inductive Circuits

The current lags the voltage by 90°, that is $\theta_v - \theta_i = 90°$, then we have:

$$p = -Q \sin 2\omega t \qquad Q = V_m I_m / 2$$

In a purely inductive circuit, the average power is zero. Therefore *no transformation of energy from electric to nonelectric form takes place*. when p is positive, energy is being stored in the magnetic fields associated with the inductive elements, and when p is negative, energy is being extracted from the magnetic fields.

To distinguish between average and reactive power, we use the units *watt* (W) for average power and **var** *(volt-amp reactive,* or VAR) for reactive power.



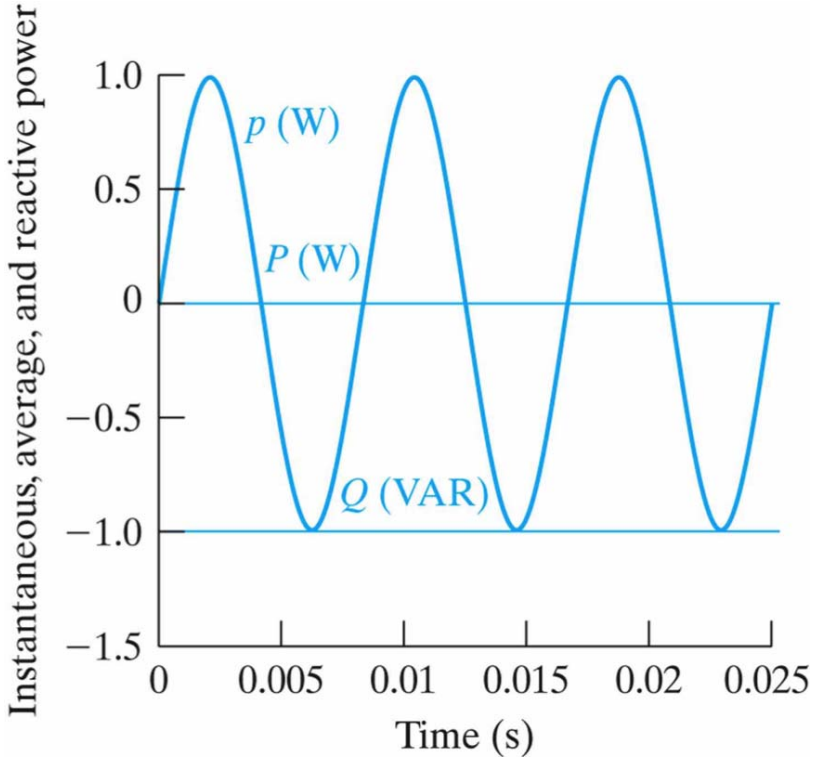
Power for Purely Capacitive Circuits

The current leads the voltage by 90°, that is $\theta_v - \theta_i = -90°$, then we have:

$$p = -Q \sin 2\omega t \qquad Q = -V_m I_m/2$$

The average power is zero, so there is no transformation of energy from electric to nonelectric form.

Note that the decision to use the current as the reference leads to Q being positive for inductors and negative for capacitors. *Power engineers recognize this difference in the algebraic sign of Q by saying that inductors demand (or absorb) magnetizing vars, and capacitors furnish (or deliver) magnetizing vars.*



The Power Factor

Power factor angle: $\theta_v - \theta_i = 90°$

Power factor: $\text{pf} = \cos(\theta_v - \theta_i)$

Reactive factor: $\text{rf} = \sin(\theta_v - \theta_i)$

Lagging power factor: current lags voltage

Leading power factor: current leads voltage

Example #1

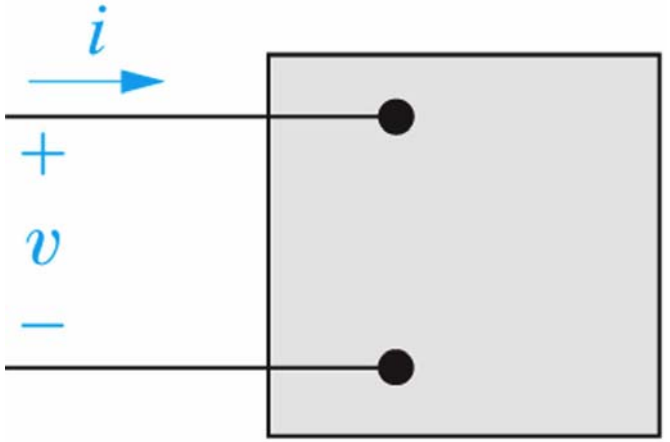
a) Calculate the average power and the reactive power at the terminals of the network shown in the Figure if

$$v = 100\cos(\omega t + 15^\circ)\text{V},$$

$$i = 4\sin(\omega t - 15^\circ)\text{ A}.$$

b) State whether the network inside the box is absorbing or delivering average power.

c) State whether the network inside the box is absorbing or supplying magnetizing vars.



Solution for Example #1

a)

$$i = 4\sin(\omega t - 15^\circ)\ \text{A} \Rightarrow i = 4\cos(\omega t - 105^\circ)\ \text{A}$$

$$P = \frac{1}{2}(100)(4)\cos[15 - (-105)] = -100\ \text{W}$$

$$Q = \frac{1}{2}100(4)\sin[15 - (-105)] = 173.21\ \text{VAR}$$

b)

The negative value of -100 W means that *the network inside the box is delivering average power to the terminals*.

c)

The passive sign convention means that, because Q is positive, *the network inside the box is absorbing magnetizing vars at its terminals*.

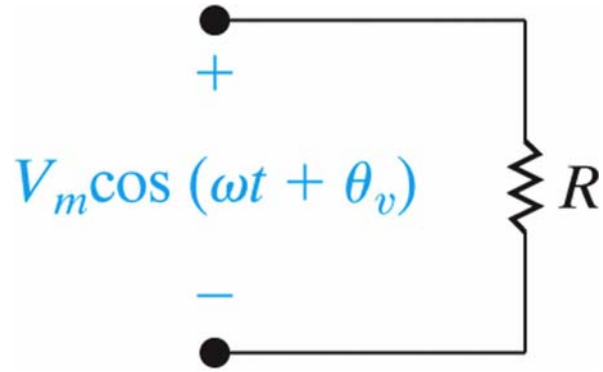
The rms Value and Power Calculations

$$P = \frac{1}{T}\int_{t_0}^{t_0+T} \frac{V_m^2 \cos^2(\omega t + \phi_v)}{R} dt$$

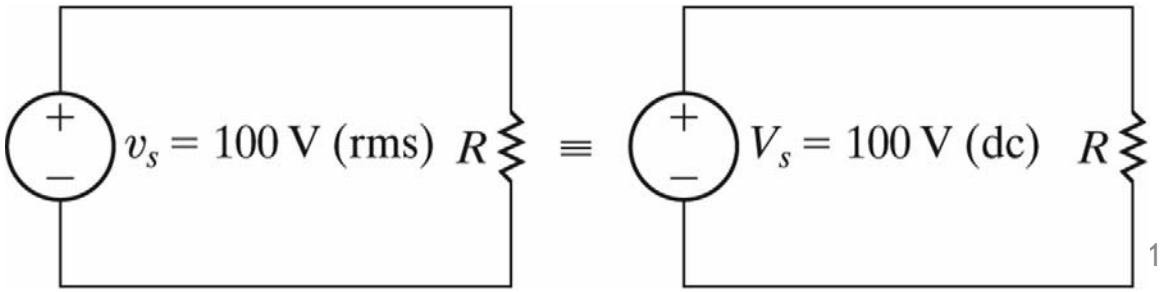
$$= \frac{1}{R}\left[\frac{1}{T}\int_{t_0}^{t_0+T} V_m^2 \cos^2(\omega t + \phi_v) dt\right]$$

$$P = \frac{V_{\text{rms}}^2}{R}$$

or $P = I_{\text{rms}}^2 R$



The rms value is also referred to as the **effective value** of the sinusoidal voltage (or current). The rms value has an interesting property: *Given an equivalent resistive load, R, and an equivalent time period, T, the rms value of a sinusoidal source delivers the same energy to R as does a dc source of the same value.*



From Eq. 10.10 (and Eq. 10.11), we have

$$P = \frac{V_m I_m}{2}\cos(\theta_v - \theta_i) = \frac{V_m}{\sqrt{2}}\frac{I_m}{\sqrt{2}}\cos(\theta_v - \theta_i) = V_{\text{eff}} I_{\text{eff}}\cos(\theta_v - \theta_i)$$

$$Q = V_{\text{eff}} I_{\text{eff}}\sin(\theta_v - \theta_i)$$

The voltage rating of residential electric wiring is often $240\ \text{V}/120\ \text{V}$ service. *These voltage levels are the rms values of the sinusoidal voltages supplied by the utility company*, which provides power at two voltage levels to accommodate low-voltage appliances (such as televisions) and higher voltage appliances (such as electric ranges).

The phasor transform of a sinusoidal function may also be expressed in terms of the rms value. The magnitude of the rms phasor is equal to the rms value of the sinusoidal function. *If a phasor is based on the rms value, we indicate this by either an explicit statement, a parenthetical "rms" adja- cent to the phasor quantity, or the subscript "eff"*.

Example #2

a) A sinusoidal voltage having a maximum ampli- tude of 625 V is applied to the terminals of a 50 Ω resistor. Find the average power delivered to the resistor.

b) Repeat (a) by first finding the current in the resistor.

Solution for Example #2

a) $V_{rms} = 625/\sqrt{2} = 441.94$ V

$$P = \frac{(441.94)^2}{50} = 3906.25 \text{ W}$$

a) $I_{rms} = V_{rms}/50 = 8.84$ A

$$P = (8.84)^2 50 = 3906.25 \text{ W}$$

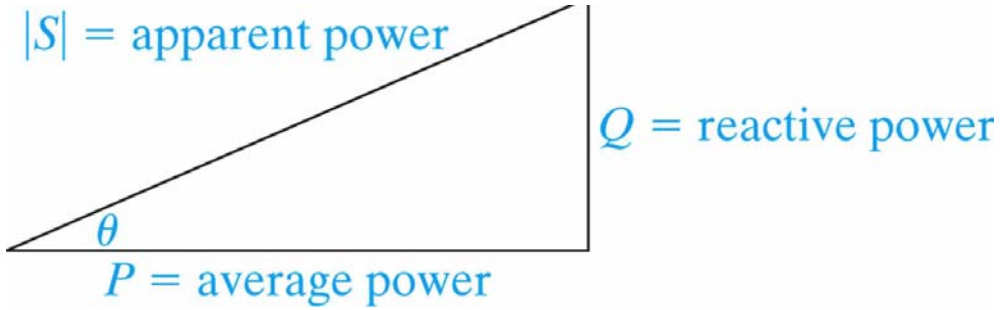
Complex Power

$$S = P + jQ$$ **complex power**

$$\tan\theta = \frac{Q}{P} = \frac{(V_m I_m/2)\sin(\theta_v - \theta_i)}{(V_m I_m/2)\cos(\theta_v - \theta_i)} = \tan(\theta_v - \theta_i)$$

$$\theta = \theta_v - \theta_i$$

Quantity	Units
Complex power	volt-amps
Average power	watts
Reactive power	var



Apparent power: the magnitude of complex power

$$|S| = \sqrt{(P^2 + Q^2)}$$

Although the average power represents the useful output of the energy-converting device*, the apparent power represents the volt-amp capacity required to supply the average power*.

Example #3

An electrical load operates at $240\ \mathrm{V}$ rms. The load absorbs an average power of $8\ \mathrm{kW}$ at a lagging power factor of 0.8.

a)Calculate the complex power of the load.

b)Calculate the impedance of the load.

Solution for Example #3

$$P = |S| \cos\theta \qquad Q = |S| \sin\theta \quad \Rightarrow \quad |S| = \frac{P}{\cos\theta} = \frac{8\ \text{kW}}{0.8} = 10\ \text{kVA}$$

$$Q = 10 \sin\theta = 6\ \text{kVAR} \quad \Rightarrow \quad S = 8 + j6\ \text{kVA}$$

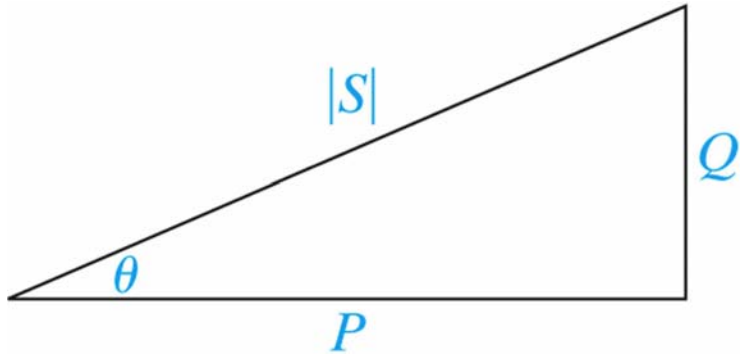
$$P = V_{\text{eff}} I_{\text{eff}} \cos(\theta_v - \theta_i) = (240) I_{\text{eff}} (0.8) = 8000\ \text{W} \quad \Rightarrow \quad I_{\text{eff}} = 41.67\ \text{A}$$

Power factor is lagging

$$\theta = \cos^{-1}(0.8) = 36.87^\circ$$

$$|Z| = \frac{|V_{\text{eff}}|}{|I_{\text{eff}}|} = \frac{240}{41.67} = 5.76$$

$$Z = 5.76 \angle 36.87^\circ\ \Omega = 4.608 + j3.456\ \Omega$$



Power Calculation

$$S = \frac{V_m I_m}{2}\cos(\theta_v - \theta_i) + j\frac{V_m I_m}{2}\sin(\theta_v - \theta_i)$$

$$= \frac{V_m I_m}{2}[\cos(\theta_v - \theta_i) + j\sin(\theta_v - \theta_i)]$$

$$= \frac{V_m I_m}{2}e^{j(\theta_v - \theta_i)} = \frac{1}{2}V_m I_m \angle(\theta_v - \theta_i)$$

$$S = V_{\text{eff}} I_{\text{eff}} \angle(\theta_v - \theta_i) = V_{\text{eff}} I_{\text{eff}} e^{j(\theta_v - \theta_i)} = V_{\text{eff}} e^{j\theta_v} I_{\text{eff}} e^{-j\theta_i} = \mathbf{V}_{\text{eff}} \mathbf{I}^*_{\text{eff}}$$

$$I_{\text{eff}} e^{-j\theta_i} = I_{\text{eff}}\cos(-\theta_i) + jI_{\text{eff}}\sin(-\theta_i)$$
$$= I_{\text{eff}}\cos(\theta_i) - jI_{\text{eff}}\sin(\theta_i) = \mathbf{I}^*_{\text{eff}}$$

$$S = \frac{1}{2}\mathbf{V}\mathbf{I}^*$$

Another Solution for Example #1

$$v = 100\cos(\omega t + 15^\circ)\ \text{V} \qquad \mathbf{V} = 100\angle 15^\circ\ \text{V}$$
$$i = 4\sin(\omega t - 15^\circ)\ \text{A} \qquad \mathbf{I} = 4\angle -105^\circ\ \text{A}$$

$$S = \frac{1}{2}\mathbf{V}\mathbf{I}^*$$

$$S = \frac{1}{2}(100\angle 15^\circ)(4\angle +105^\circ) = 200\angle 120^\circ = -100 + j173.21\ \text{VA}$$

$$S = P + jQ$$

$$P = -100\ \text{W}$$
$$Q = 173.21\ \text{VAR}$$

Alternate Forms for Complex Power

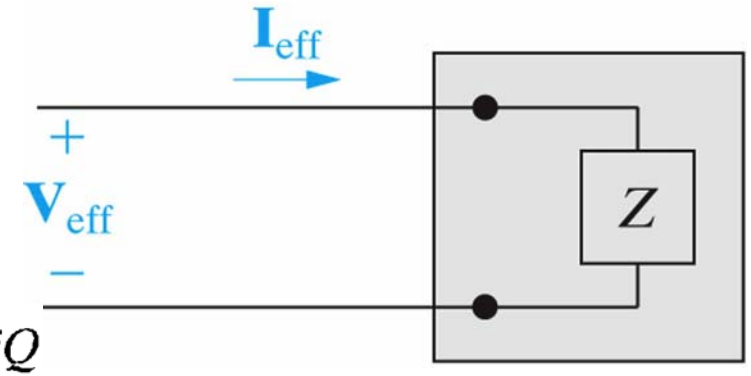
$$\mathbf{V}_{\text{eff}} = Z\mathbf{I}_{\text{eff}}$$

$$S = \mathbf{V}_{\text{eff}}\mathbf{I}^*_{\text{eff}}$$

$$S = Z\mathbf{I}_{\text{eff}}\mathbf{I}^*_{\text{eff}}$$

$$= |\mathbf{I}_{\text{eff}}|^2(R + jX)$$

$$= |\mathbf{I}_{\text{eff}}|^2R + j|\mathbf{I}_{\text{eff}}|^2X = P + jQ$$



$$P = |\mathbf{I}_{\text{eff}}|^2R = \frac{1}{2}I_m^2R$$

$$Q = |\mathbf{I}_{\text{eff}}|^2X = \frac{1}{2}I_m^2X$$

$$S = \mathbf{V}_{\text{eff}}\left(\frac{\mathbf{V}_{\text{eff}}}{Z}\right)^* = \frac{|\mathbf{V}_{\text{eff}}|^2}{Z^*} = P + jQ$$

$$P = \frac{|\mathbf{V}_{\text{eff}}|^2}{R}$$

$$Q = \frac{|\mathbf{V}_{\text{eff}}|^2}{X}$$

Example #4

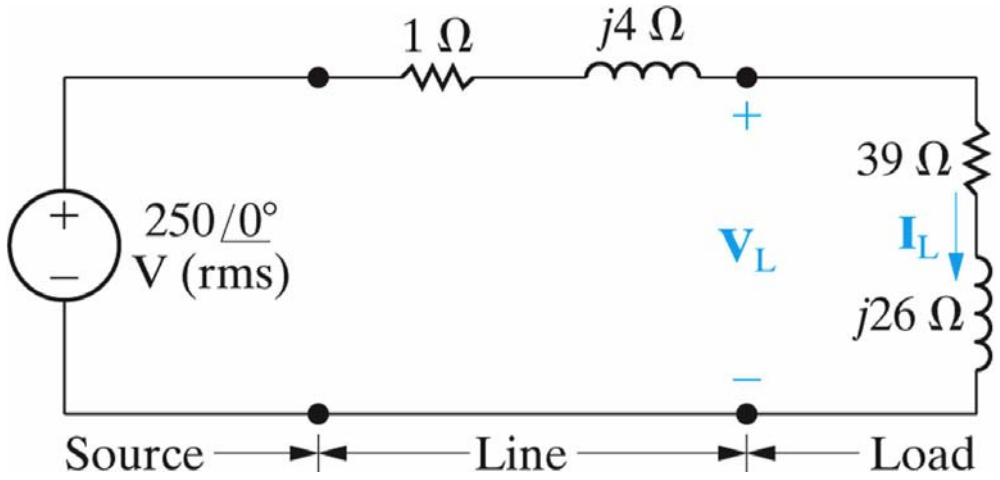
In the given circuit, a load having an impedance of $39 + j26\ \Omega$ is fed from a voltage source through a line having an impedance of $1 + j4\ \Omega$. The effective, or rms, value of the source voltage is $250\ \text{V}$.

a)Calculate the load current $\mathbf{I}_\text{L}$ and voltage $\mathbf{V}_\text{L}$.

b)Calculate the average and reactive power delivered to the load.

c)Calculate the average and reactive power delivered to the line.

d)Calculate the average and reactive power supplied by the source.



Solution for Example #4

a) $$\mathbf{I}_L = \frac{250\ \angle 0°}{40 + j30} = 4 - j3 = 5\ \angle -36.87°\ \text{A (rms)}$$

$$\mathbf{V}_L = (39 + j26)\mathbf{I}_L = 234 - j13 = 234.36\ \angle -3.18°\ \text{V (rms)}$$

$$S = \mathbf{V}_L\mathbf{I}_L^* = (234 - j13)(4 + j3) = 975 + j650\ \text{VA}$$ absorb

b) $P = (5)^2(1) = 25\ \text{W}$

$Q = (5)^2(4) = 100\ \text{VAR}$

absorb

c) $S = 25 + j100 + 975 + j650$

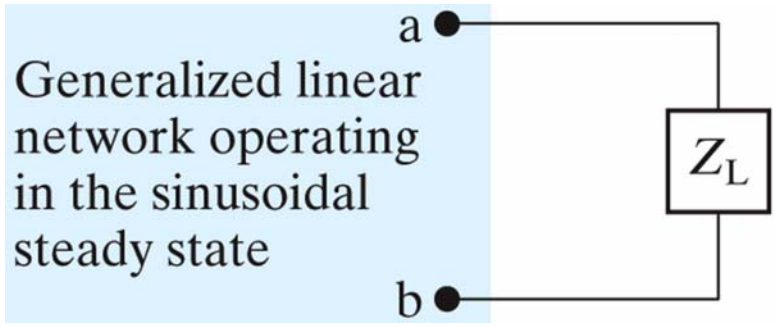
$= 1000 + j750\ \text{VA}$

delivered by source

Maximum Power Transfer

Maximum average power transfer:

$$Z_L = Z_{Th}^*$$



$$Z_{Th} = R_{Th} + jX_{Th} \qquad Z_L = R_L + jX_L$$

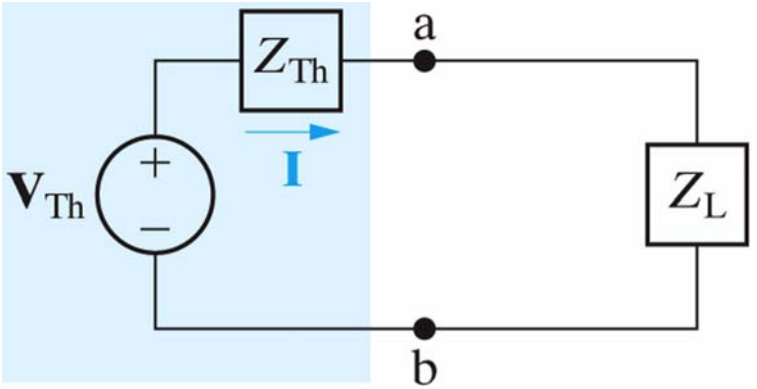
$$\mathbf{I} = \frac{\mathbf{V}_{Th}}{(R_{Th} + R_L) + j(X_{Th} + X_L)}$$

Thévenin equivalent

$$P = |\mathbf{I}|^2 R_L$$

$$P = \frac{|\mathbf{V}_{Th}|^2 R_L}{(R_{Th} + R_L)^2 + (X_{Th} + X_L)^2}$$

fixed quantities



$$\frac{\partial P}{\partial X_L} = \frac{-|\mathbf{V}_{Th}|^2 2R_L(X_L + X_{Th})}{[(R_L + R_{Th})^2 + (X_L + X_{Th})^2]^2}$$

$$\frac{\partial P}{\partial R_L} = \frac{|\mathbf{V}_{Th}|^2[(R_L + R_{Th})^2 + (X_L + X_{Th})^2 - 2R_L(R_L + R_{Th})]}{[(R_L + R_{Th})^2 + (X_L + X_{Th})^2]^2}$$

$$X_L = -X_{Th} \qquad R_L = \sqrt{R_{Th}^2 + (X_L + X_{Th})^2} = R_{Th}$$

$$\boxed{Z_L = Z_{Th}^*}$$

$$P_{max} = \frac{|\mathbf{V}_{Th}|^2 R_L}{4R_L^2} = \frac{1}{4}\frac{|\mathbf{V}_{Th}|^2}{R_L}$$

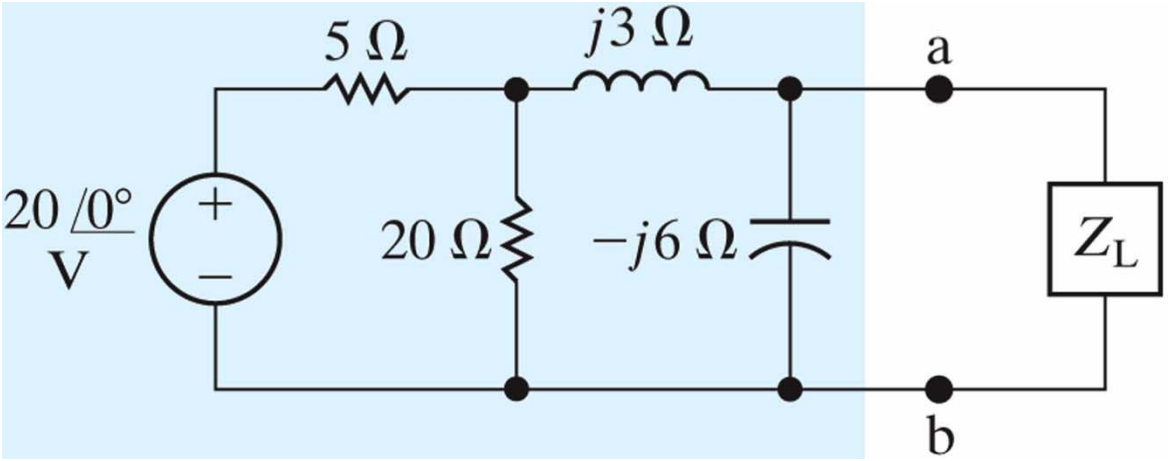
$$P_{max} = \frac{1}{8}\frac{\mathbf{V}_m^2}{R_L}$$

What if R_L and X_L are restricted?

Example #5

a) For the given circuit, determine the impedance Z_L that results in maximum average power transferred to Z_L.

b) What is the maximum average power transferred to the load impedance determined in (a)?



Solution for Example #5

a)

$$\mathbf{V}_{Th} = \frac{16\ \angle 0°}{4 + j3 - j6}(-j6)$$

$$= 19.2\ \angle -53.13° = 11.52 - j15.36\ \text{V}$$

$$Z_{Th} = \frac{(-j6)(4 + j3)}{4 + j3 - j6} = 5.76 - j1.68\ \Omega$$

$$Z_L = Z_{Th}^*$$

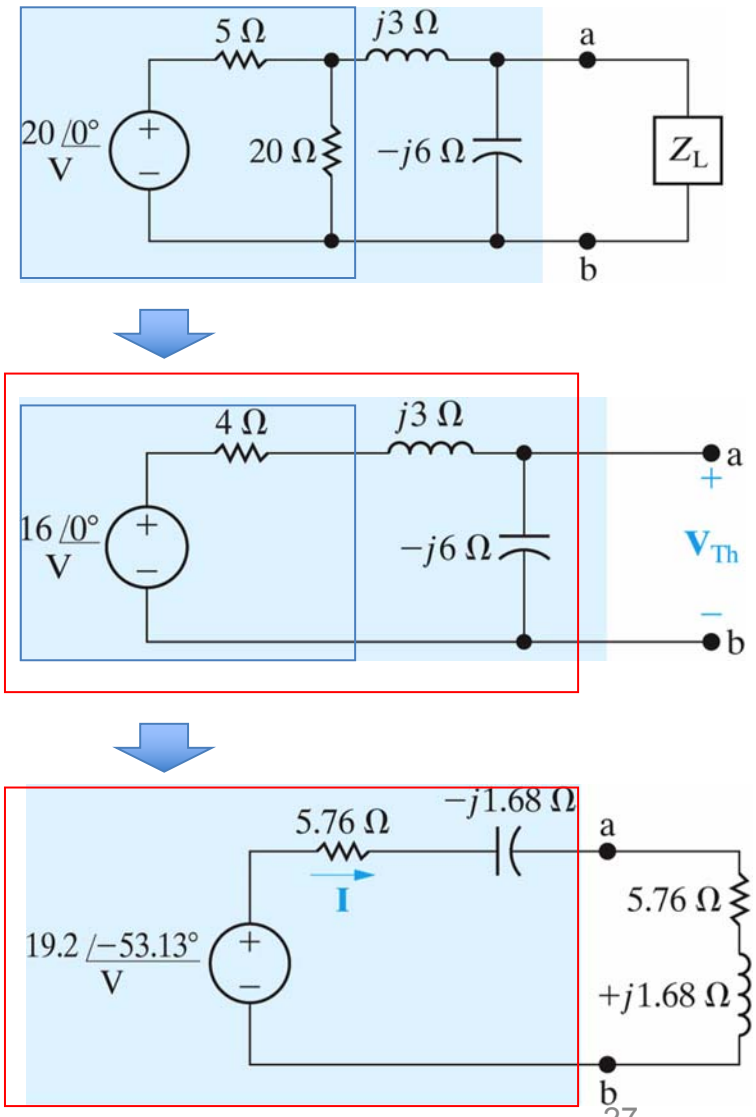
b)

$$Z_L = 5.76 + j1.68\ \Omega \qquad I_{eff} = \frac{19.2/\sqrt{2}}{2(5.76)} = 1.1785\ \text{A}$$

$$P = |\mathbf{I}_{eff}|^2 R$$

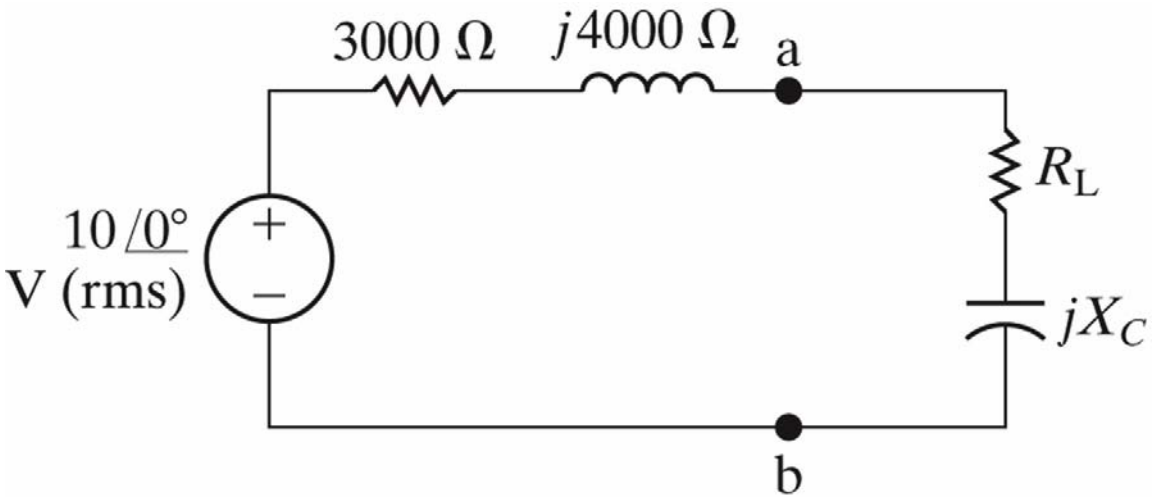
$$P = I_{eff}^2(5.76) = 8\ \text{W}$$

$$P_{max} = \frac{1}{8}\frac{\mathbf{V}_m^2}{R_L}$$



Example #6

Assume that the load resistance can be varied between 0 and $4000\ \Omega$ and that the capacitive reactance of the load can be varied between 0 and $-2000\ \Omega$. What settings of R_{L} and X_{L} transfer the most average power to the load? What is the maximum average power that can be transferred under these restrictions?



Solution for Example #6

$$P = \frac{|\mathbf{V}_{Th}|^2 R_L}{(R_{Th} + R_L)^2 + (X_{Th} + X_L)^2}$$

X_L as close to $-X_{Th}$ as possible

$$X_L = -2000\ \Omega$$

$$\partial P/\partial R_L \longrightarrow 0 \quad \Rightarrow \quad R_L \longrightarrow \sqrt{R_{Th}^2 + (X_L + X_{Th})^2}$$

$$R_L = \sqrt{3000^2 + (-2000 + 4000)^2} = 3605.55\ \Omega$$

$$Z_L = 3605.55 - j2000\ \Omega$$

$$\mathbf{I}_{eff} = \frac{10\ \angle 0°}{6605.55 + j2000} = 1.4489\ \angle -16.85°\ \text{mA}$$

$$P = (1.4489 \times 10^{-3})^2 (3605.55) = 7.57\ \text{mW}$$

Summary

- Instantaneous power
- Average/reactive power
- Power/reactive factor
- Complex/apparent power
- Maximum power transfer